MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 8316

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, integral cavity, mechanically tuned, 5,430 to 5,930 MHz, waveguide output, 1.2 W.

ABSOLUTE RATINGS:

| Parameter: | Ef | Ers | Ik | Pi | Er | T (cathode seal) | T (body) | Tuner-torque |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | mAdc | W | Vdc | °C | °C | in. -lbs |
| Maximum: | 6.9 | 900 | 110 | 99 | -1,100 | 100 | 120 | 6 |
| Minimum: | 6.0 | --- | --- | --- | -25 | --- | --- | --- |
| | | | | | | (see figures 1 and 2) | (see figures 1 and 2) | |

PHYSICAL CHARACTERISTICS:

| | | | |
|---|---|---|---|
| Dimensions: | See figures 1 and 2 | RF connections: | Waveguide, mates with UG-148C/U (MIL-F-3922/62), UG-149A/U (MIL-F-3922/57). |
| Cathode: | Unipotential, oxide coated | | |
| Mounting position: | Any | | |
| Cooling: | Forced air (see note 2) | Base: | Shortened, medium shell, octal, 8-pin, low-loss phenolic. |
| Cap: | Small (C1-1) | | |

TEST CONDITIONS:

| Parameter: | Ef | Ehk | Ers | Er | T | VSWR | F |
|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | °C | σ' | MHz |
| | 6.3 | 0 | 900 | -250 to -550 | Note 1 | 1.1:1 (max) | F2 |

| Frequency | | |
|---|---|---|
| F | MHz | ± |
| F1 | 5,430 | 6 |
| F2 | 5,680 | 6 |
| F3 | 5,930 | 6 |

GENERAL:

*Qualification* - Required



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| METHOD | REQUIREMENT OR TEST | NOTES | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|
| | | | | | MINIMUM | MAXIMUM | |
| | Qualification inspection | | | | | | |
| --- | Shelf life | 2 | | --- | --- | --- | --- |
| --- | Shelf life end point: | | | | | | |
| 4250 | Power output | - | F3; Er = -410 to -550 Vdc | Po | 1.2 | --- | W |
| 4280 | Electronic tuning range | 3,4 | F1 to F3 | ΔF | 35 | --- | MHz |
| 4215 | Reflector voltage coefficient | 4 | F = F1, F2, F3 | ΔF/Er | 0.250 | 0.650 | MHz/V |
| 4027 | Temperature coefficient | 4,5,6 | | ΔF/ΔT | 0 | 0.06 | MHz/°C |
| 1031 | High-frequency vibration | 7 | Ers = 900 Vdc | Ir | --- | 10 | μAdc |
| 1031 | Sweep-frequency vibration | 8,9 | F = 25 to 2,000 Hz; 10G; t = 5 minutes | ΔF | --- | ±5 | MHz |
| 4223 | Mechanical tuning fatique | 8 | Ef = 6.3 V | --- | 50 | --- | Cycles |
| 1042 | Shock | 8 | Test condition C | --- | --- | --- | --- |
| | Quality conformance inspection, part 1 | 10 | | | | | |
| 1201 | Permanent shorts | - | Ers = 0 | --- | --- | --- | --- |
| 4289 | Heater current (nonoperating) | - | | If | 0.70 | 0.90 | A |
| 4270 | Heater-cathode leakage | - | Ef = 0; Ehk = 45 Vdc | Ihk | 0 | 150 | μAdc |
| 4250 | Power output | 4,5 | F = F1, F2, F3 | Po | 1.2 | --- | W |
| 4213 | Reflector voltage (1)<br>(2)<br>(3) | 4<br>4<br>4 | F1<br>F2<br>F3 | Er<br>Er<br>Er | -250<br>-340<br>-410 | -350<br>-430<br>-550 | Vdc<br>Vdc<br>Vdc |
| 4229 | Total reflector current | 11,12 | Er = -200 to -300 Vdc | Ir | 0 | 5 | μAdc |
| | Quality conformance inspection, part 2 | | | | | | |
| 4266 | Direct-interelectrode capacitance | - | Reflector to shell | C | --- | 5 | pF |
| 4223 | Operating torque or force | - | | Torque | --- | 2.5 | in. -lb |
| 4223 | Tuner stop endurance | - | | Torque | 6 | --- | in. -lb |
| 4223 | Mechanical tuning range | 4, 13, 14 | | F | 5,430 | 5,930 | MHz |
| 4223 | Resettability | 4,13,14 | F = 5,430 to 5,930 MHz | --- | 30 | 70 | MHz/turn |
| 1256 | Electrode current (cathode) | 11 | Er = -200 to -300 Vdc | Ik | --- | 110 | mAdc |
| 4214 | Cathode emission | 15 | Ef = 5.7 V; t = 60 sec; Er =-200 to -300 Vdc | ΔIk/Ik | --- | 10 | % |
| 1105 | Permanence of marking | - | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | |
| --- | Life test | 16 | Group D | t | 500 | --- | hrs |
| --- | Life-test end point: | | | | | | |
| 4250 | Power output | 4,5 | F = F1, F2, F3 | Po | 0.96 | --- | W |



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NOTES:

1. Sufficient cooling shall be distributed over the the entire tube so that the absolute ratings of T (cathode seal) and T (body) are not exceeded. Recommended cooling at sea level, when operating at maximum power input (Pi), is 35 cfm.

2. Storage. The tube shall be capable of storage for 12-months in shipping containers at air temperatures within shipping containers from minus 55°C to plus 70°C without impairment of subsequent performance. This test shall not preclude provisional qualification approval. The tube shall withstand this period of storage without requiring any maintenance. Furthermore, the tube shall be capable of operating in a manner to meet all requirements specified herein after 50 minutes of initial warmup. The above shall be demonstrated by means of the following tests:

   a. Cold cycle. With the tube in the test chamber and beginning at laboratory ambient temperature, the temperature of the chamber shall be cycled from plus 20°C to minus 55°C and back to plus 20°C. This cycle shall be repeated 5 times. The minus 55°C temperature shall be maintained for at least 3 hours during each cycle. Also, the plus 20°C temperature shall be maintained for at least 3 hours between cycles. After the completion of the required cycles the tube shall be stabilized at the ambient laboratory temperature and then tested for satisfactory performance.

   b. Hot cycle. Starting at laboratory ambient temperature, the temperature of the chamber shall be cycled from plus 20°C to plus 70°C and back to plus 20°C. This cycle shall be repeated 5 times. The plus 70°C temperature shall be maintained for at least 3 hours between cycles. At the conclusion of the last cycle the tube shall be stabilized at ambient laboratory temperature and then checked for satisfactory performance. Either the hot or cold cycle may be performed first. The rate of temperature change of the test chanber from plus 20°C to either minus 55°C or plus 70°C shall be accomplished in no more than 3 hours. Likewise, the change from plus 70°C or minus 55°C to plus 20°C shall require no more than 3 hours.

3. The electronic tuning bandwidth is the frequency difference between the half-power points of the reflector voltage modulation characteristic with the tube tuned in accordance with notes 4 and 5.

4. The reflector voltage shall be adjusted to that value within the range specified which results in maximum power output from the tube under test.

5. The output power shall be measured with appropriate waveguide equipment. This equipment shall include a suitable impedance transformer, a directional coupler, and a termination whose VSWR is less than 1.1 to 1. Power shall be measured on the branch arm of the directional coupler by a bolometer element and an appropriate wattmeter bridge.

6. The initial frequency shall be allowed to stabilize by operating at practically constant ambient temperature for at least 10 minutes, then temperature cycled once through the temperature range used before actual measurements are made. The temperature range for the test shall be any convenient 60°C rise within the range minus 40°C to plus 100°C. The rate of change of ambient temperature shall not be greater than 2°C per minute. During the test, the tube shall be operated in a rectangular enclosure that has a base at least 1-foot square and at least 20-inches high. The air within the enclosure shall be circulated with an airblast directed on the tube. All specified voltages shall be maintained constant during this test.

7. The reflector current shall be recorded with an instrument whose frequency response is from dc to 100 Hz with no more than 3 dB fall-off at 100 Hz. There shall be no reflector current bursts greater than the limits specified herein.

8. After completing this test, the TUT shall satisfy the requirement of the power output test.

9. The specified voltages shall be applied to the tube during vibration. The difference of frequency change shall be measured with a suitable device. The value of vibration frequency change $\Delta F$ shall not exceed the limit specified herein at any point in the swept frequency during the last complete cycle of cycling vibration.

10. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

11. The tube shall not be oscillating during the test. If the tube oscillates with the specified voltages applied, adjust Er or detune the resonator until the oscillation ceases.



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NOTES: -Continued

12. Apply heater voltage for 10 minutes before application of other specified voltages. Observe reflector current during first 30 seconds after application of beam voltage and record current or continue observation as indicated in table I.

TABLE I. Reflector current at end of 30-second interval.

| Magnitude | Trend | Disposition |
|---|---|---|
| 3 µA or less | Steady or decreasing | Record 30-second reading. |
| 3 µA or less | Increasing | Record maximum reading within 2-minute interval after application of beam voltage. |
| 3 µA or more | Steady or decreasing | Record 30-second reading. |
| 3 µA or more | Increasing | Record maximum reading within 2-minute interval after application of beam voltage. |

If recorded reading is less than 3 µA, the tube meets the requirements specified herein.

13. Clockwise rotations of the tuner (as viewed from the reflector end of the tube) will increase the frequency of the resonator.

14. The specified tuning range of the resonator will be covered by 10 ±3 turns of the tuning strut.

15. The heater voltage shall be decreased from 6.3 to 5.7. The percentage change in cathode current shall not exceed the specified limit when determined as follows:

$$\frac{\Delta Ik}{Ik}\% = \frac{Ik\ (\text{with } Ef = 6.3\ V) - Ik\ (\text{with } Ef = 5.7\ V)}{Ik\ (\text{with } Ef = 6.3\ V)}$$

16. The tube shall be cycled in accordance with the following schedule: (Minimum life refers to "on" cycle only.)

| Cycle | Ef | Er | Ers | Duration (minutes) |
|---|---|---|---|---|
| Off | 6.3 V | -375 Vdc | 0 Vdc | 25 |
| On | 6.3 V | -375 Vdc | 900 Vdc | 15 |

17. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

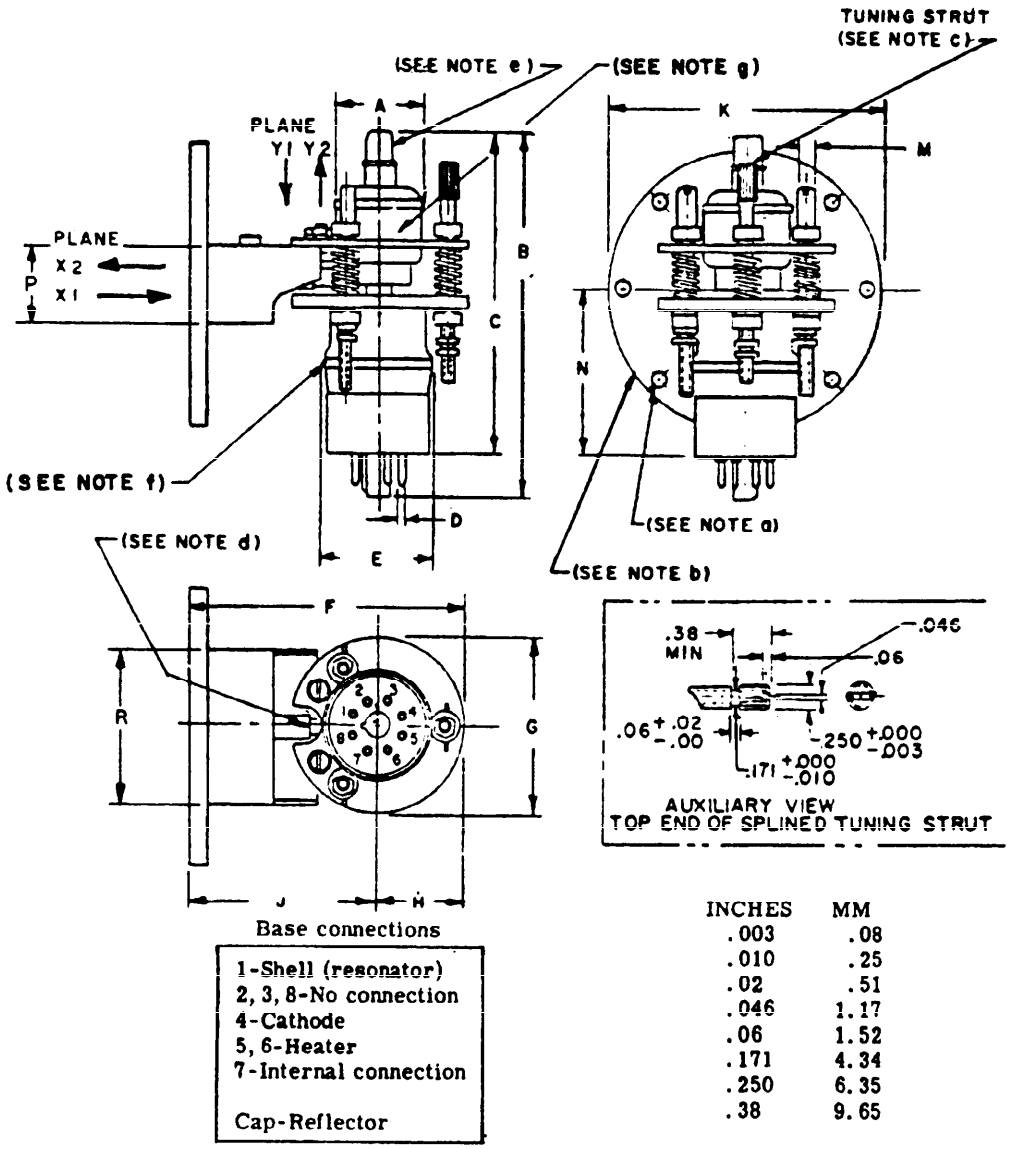
Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3191)



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Base connections

| |
|---|
| 1-Shell (resonator) |
| 2, 3, 8-No connection |
| 4-Cathode |
| 5, 6-Heater |
| 7-Internal connection |
| |
| Cap-Reflector |

| INCHES | MM |
|---|---|
| .003 | .08 |
| .010 | .25 |
| .02 | .51 |
| .046 | 1.17 |
| .06 | 1.52 |
| .171 | 4.34 |
| .250 | 6.35 |
| .38 | 9.65 |

FIGURE 1. Outline drawing of electron tube type 8316.



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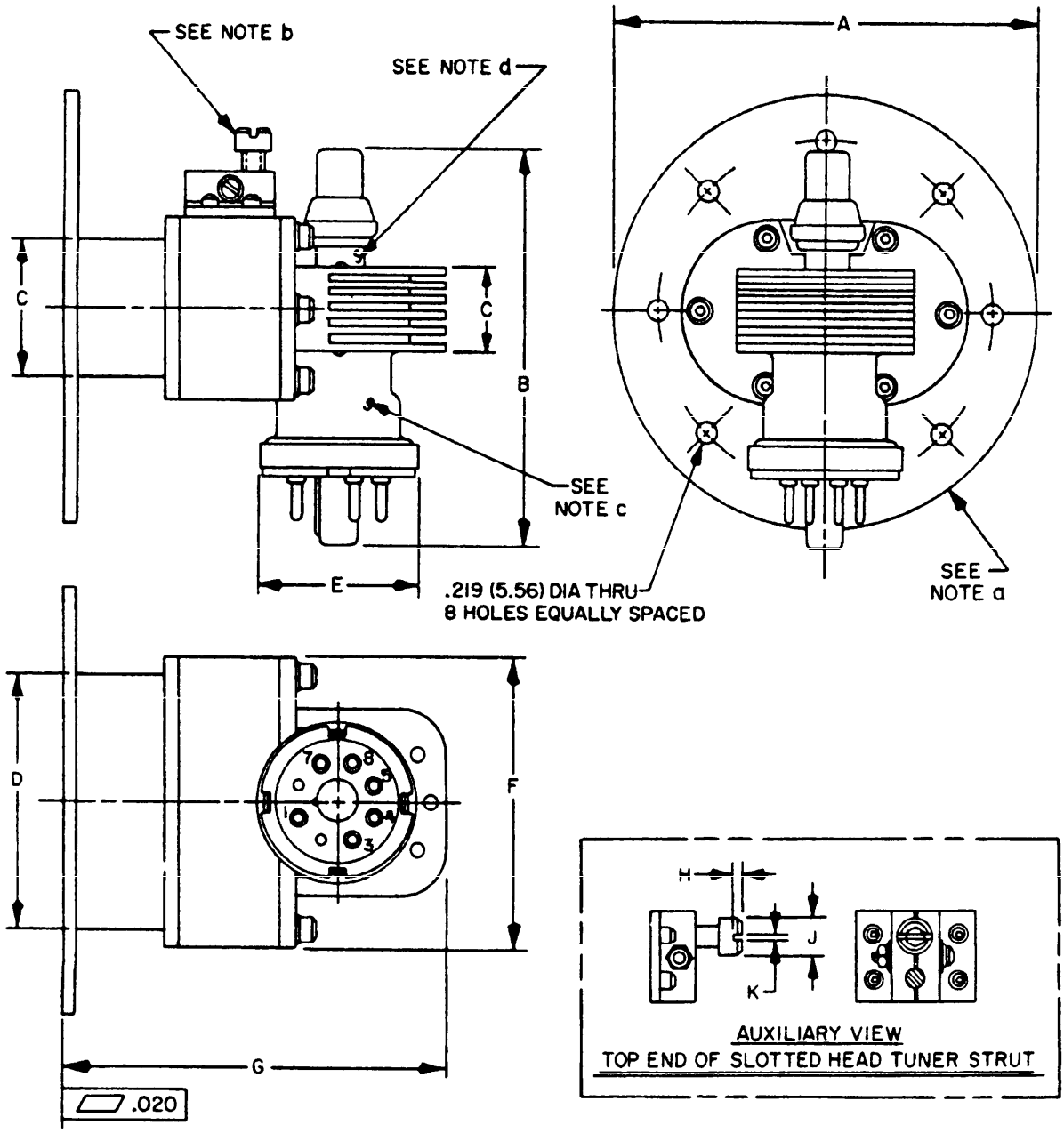
| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | 1.187 (30.15) | 1.219 (30.96) |
| B | | 4.750 (120.65) |
| C | | 4.188 (106.38) |
| D | | .098 (2.49) |
| E | 1.437 (36.50) | 1.563 (39.70) |
| F | | 4.078 (103.58) |
| G | 2.313 (58.75) | 2.375 (60.33) |
| H | 1.156 (29.36) | 1.188 (30.18) |
| J | | 2.875 (73.03) |
| K | 3.594 (91.29) | 3.656 (92.86) |
| M | .234 DIA (5.94) | .266 DIA (6.76) |
| N | | 2.188 (55.58) |
| P | | 1.190 (30.23) |
| R | | 2.180 (55.37) |

NOTES:
a. Eight holes 7/32 diameter through (equally spaced).
b. Mates with AN type UG-148C/U (MIL-F-3922/62). UG-149A/U (MIL-F-3922/57).
c. Involute spline data: American Standard B92.1-1970 class A fit, 11 teeth, 48/96 diametral pitch, 30° pressure angle.
d. The connection between the waveguide and the antenna shall be airtight.
e. Small cap square with axis of the tube within .025 (.64 mm).
f. Measure the temperature of the cathode seal at this point.
g. Measure the temperature of the body at this point.

FIGURE 1. Outline drawing of electron tube type 8316 - Continued.



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| PIN | CONNECTION |
|---|---|
| PIN 1 | SHELL (RESONATOR) |
| PIN 3 | NO CONNECTION |
| PIN 4 | CATHODE |
| PIN 5 | HEATER |
| PIN 6 | HEATER |
| PIN 7 | NO CONNECTION |
| CAP | REFLECTOR |

FIGURE 2. Alternate outline drawing of electron tube type 8316.



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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses. | |
|---|---|---|
| | MINIMUM | MAXIMUM |
| Quality conformance inspection, part 2 | | |
| A | 3.594 (91.29) | 3.656 (92.86) |
| B | | 4.750 (120.65) |
| C | | 1.190 (30.23) |
| D | | 2.180 (55.37) |
| E | 1.250 (31.75) | 1.374 (34.90) |
| F | | 2.450 (62.23) |
| G | | 3.330 (84.58) |
| H | .042 (1.07) | .082 (2.08) |
| J | .292 (7.42) | .332 (8.43) |
| K | .020 (.51) | .060 (1.52) |

NOTES:
a. Mates with "AN" type UG-148C/U (MIL-F-3922/62), UG-149A/U (MIL-F-3922/57).
b. Slotted socket head tuner strut.
c. Measure temperature of cathode here.
d. Measure temperature of body here.

FIGURE 2. Alternate outline drawing of electron tube type 8316 - Continued.



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